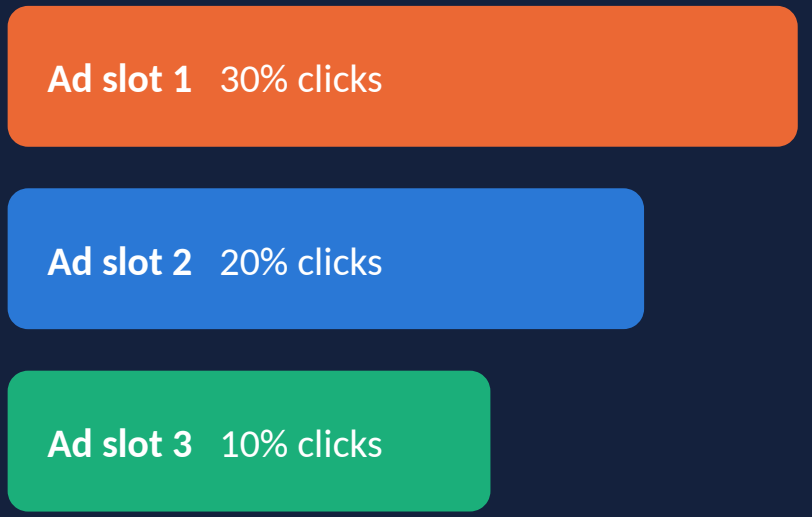
Game-Theoretic Analysis of Online Advertising Competition

Game Theory course project · B.Tech CSE · Manipal University Jaipur



Who gets the top slot, and who pays what?

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THE PROBLEM

Online advertising is a strategic game

Shared customers

When one firm advertises more, it takes customers from its rivals. Its gain depends on what they spend.

Auctions set the price

There is no fixed price for an ad. The auction rules and rivals' bids decide what you pay.

Played every day

Budgets and bids are revised daily, so firms watch and react to each other.

PROBLEM STATEMENT

How do competing firms choose ad budgets and bids, what outcome does this competition reach, and how do the platform's auction rules change who gains and who loses?

OBJECTIVES AND APPROACH

Five models of one market

M1

Advertising war

Low or High budget?

Prisoner's Dilemma

M1b

Repeated game

Same war, every month

Trigger strategies

M2

Budget contest

Any budget, n firms

Tullock contest

M3

Keyword auctions

GSP vs VCG vs GFP

Mechanism design

M4 + M5

Dynamics and display ads

Bidding over time, bid shading, reserve price

Bayesian equilibrium

SAME FOUR STEPS FOR EVERY MODEL

1. Define the game: players, strategies, payoffs
2. Solve by theory: the Nash equilibrium formula
3. Solve by computer: search or let players adapt
4. Compare and vary the inputs

Both firms overspend: a Prisoner's Dilemma

Advertising war: payoff matrix (₹ lakh profit)

	Firm B: Low	Firm B: High
Firm A: Low	40.0 , 40.0 best joint profit	15.0 , 45.0
Firm A: High	45.0 , 15.0	**20.0 , 20.0** Nash equilibrium

profit = market × share − spend, share = own spend ÷ total spend

High is dominant: 45 > 40 if the rival is Low, 20 > 15 if it is High

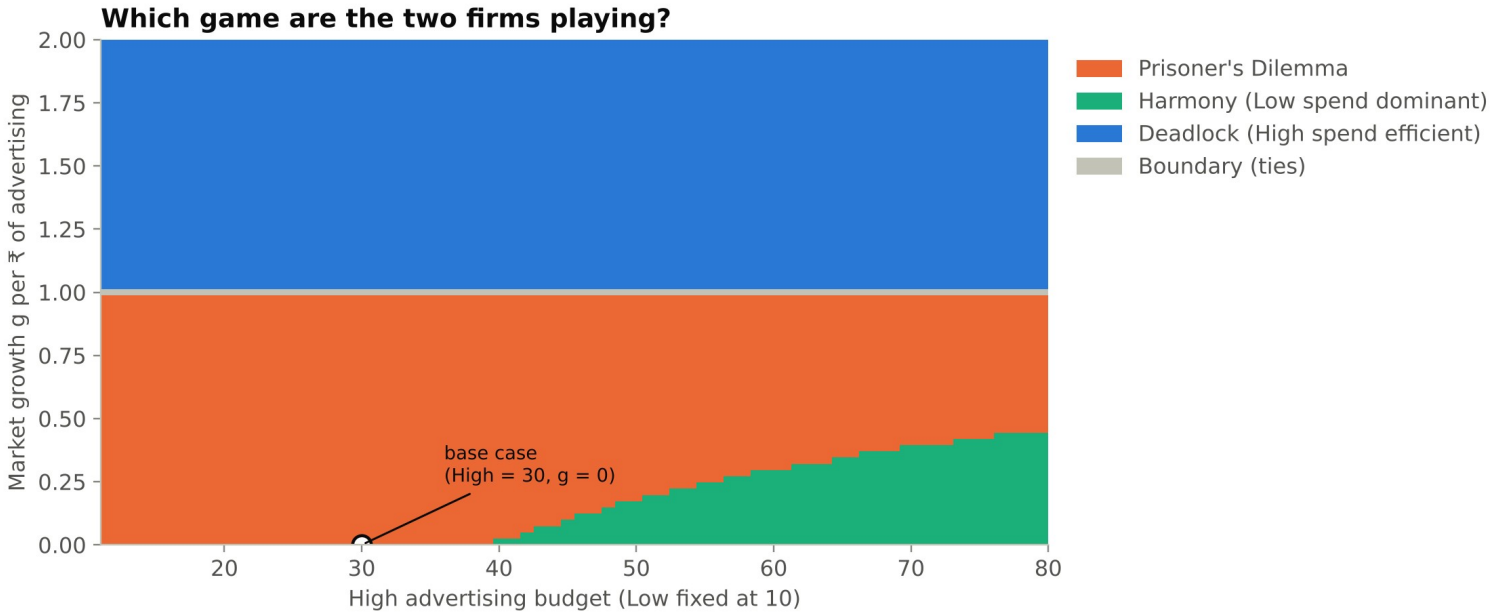
Only equilibrium: (High, High) at ₹20 lakh profit each

Both would earn ₹40 lakh at (Low, Low), but neither can trust the other

50%

of joint profit is lost at the equilibrium. The extra ₹40 lakh goes to the ad platform.

Overspending happens when ads steal customers



We solved 5,670 versions of the game, changing the High budget (11 to 80) and how much each rupee of ads grows the market (g from 0 to 2).

41%

Prisoner's Dilemma

g < 1: ads mostly steal customers

49%

Deadlock

g > 1: ads create new demand, so High is good for both

8%

Harmony

High budget too big for the market: Low wins

With 3 levels (10 / 25 / 40) the unique equilibrium is ₹25 lakh each, exactly what Model 2 predicts.

Playing every month makes low budgets possible

Cooperation (Low, Low) survives if

$$\delta \geq 0.20$$

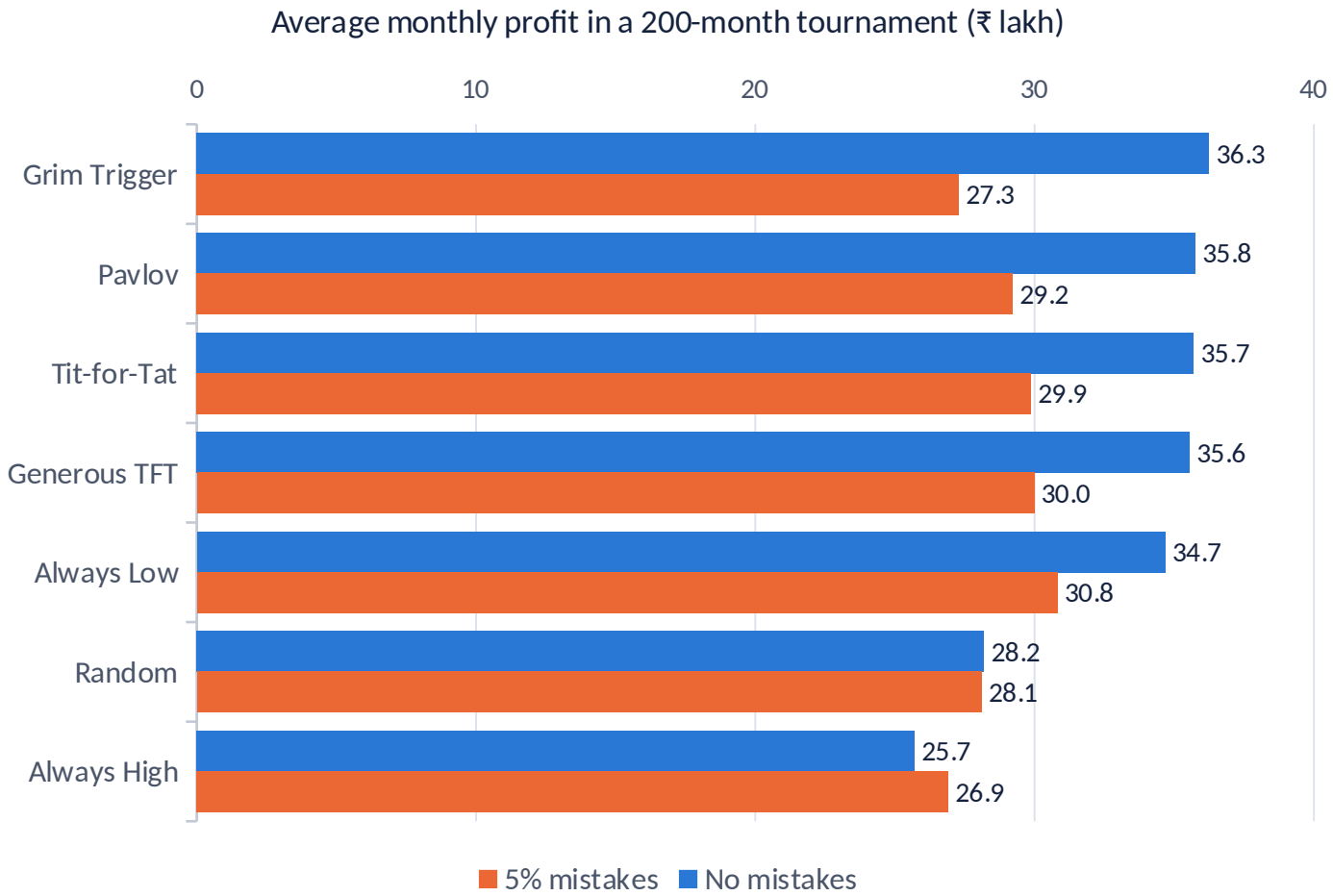
$\delta^* = (T - R) \div (T - P) = (45 - 40) \div (45 - 20)$

δ (delta) = how much a firm values next month's profit

Reciprocal strategies win: Grim Trigger, Pavlov and Tit-for-Tat earn about ₹36 lakh a month, close to the cooperative ₹40

Always High comes last with no mistakes

With 5% mistakes Grim Trigger falls to 6th: one error starts a war that never ends. Forgiveness matters.



Any budget, any number of firms

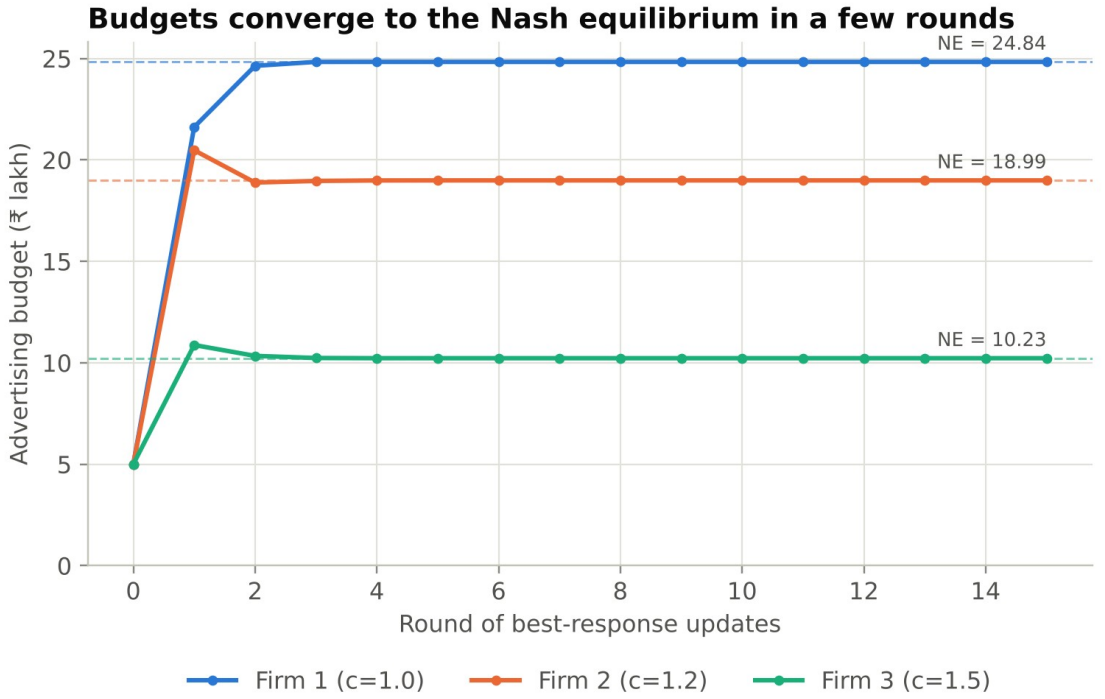
Firm i's profit

$$\pi_i = V \times x_i \div (x_1 + \ldots + x_n) - c_i \times x_i$$

V = value of the customers (₹100 lakh), x_i = budget, c_i = cost per ₹ of ads

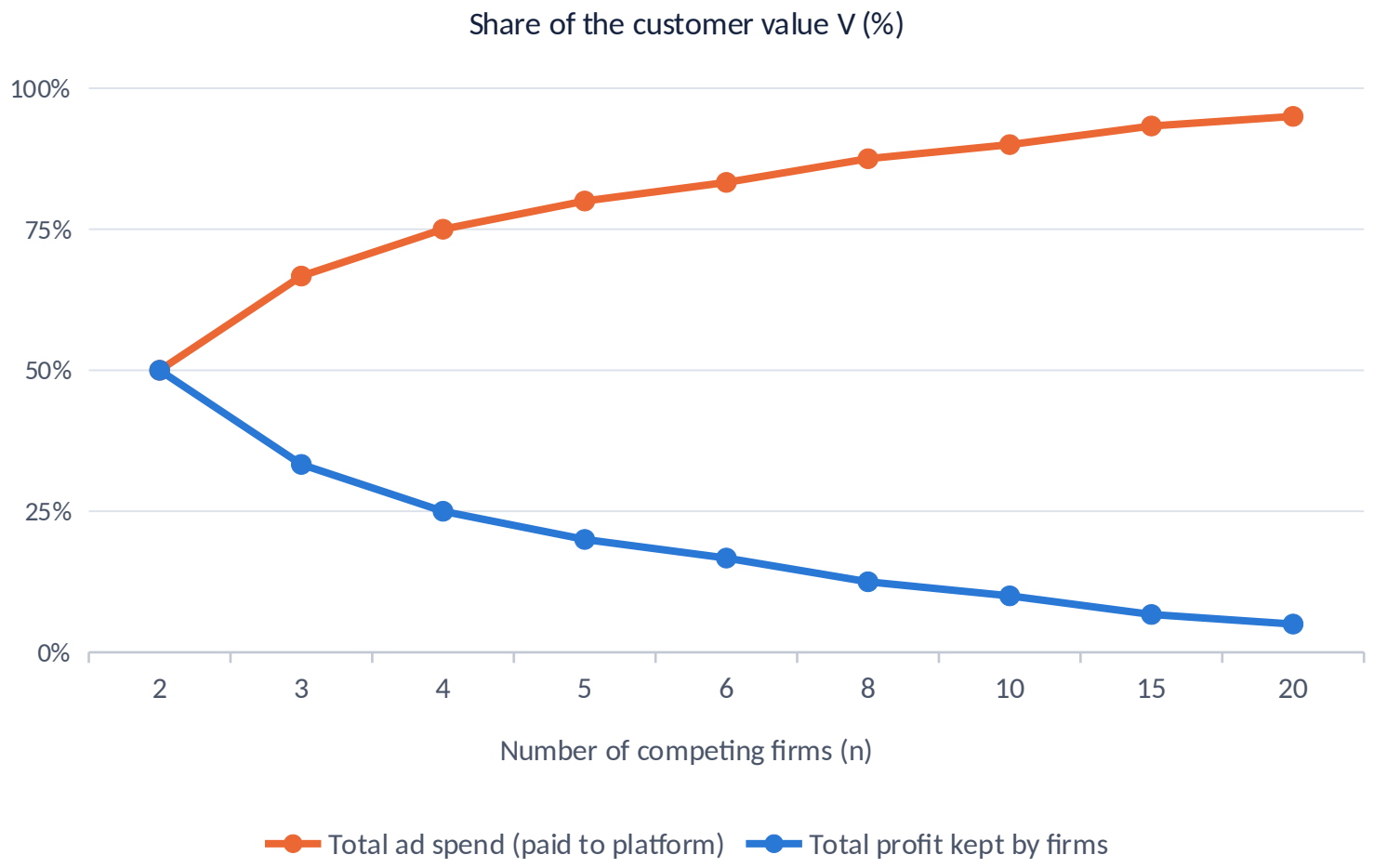
Firm	Cost	NE budget	Share	Profit
Firm 1	1.0	24.84	45.9%	21.11
Firm 2	1.2	18.99	35.1%	12.34
Firm 3	1.5	10.23	18.9%	3.58

Raise Firm 3's cost to 3.0: its budget falls to 0. The weakest firm stops advertising.



Firms take turns choosing their best reply. Budgets reach the equilibrium formula within 3 rounds.

More rivals, more value goes to the platform



Total spend = (n − 1) ÷ n × V

50% of the value goes to the platform with 2 firms

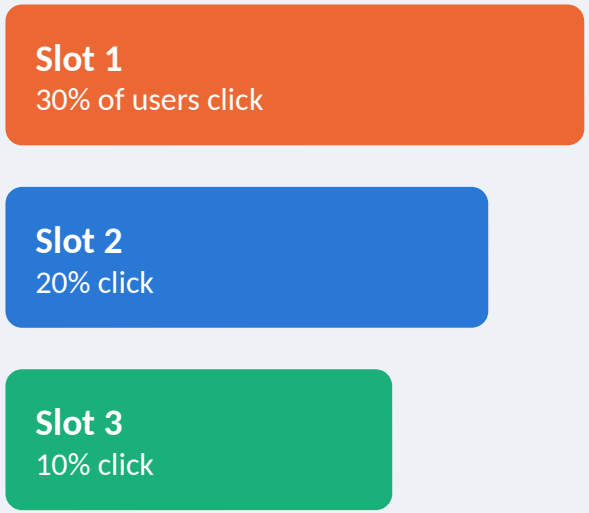
90% of the value goes to the platform with 10 firms

95% of the value goes to the platform with 20 firms

Profit per firm falls from ₹25 lakh (2 firms) to ₹1 lakh (10 firms).

Three ways to price an ad slot

Search: "running shoes"



Advertisers bid per click. Ads are ranked by bid (× quality score on Google).

GSP

Generalized Second Price

PAYS

Pay the next-highest bid

USED BY

Google Ads, Microsoft Ads

TRUTHFUL BIDDING BEST?

No

VCG

Vickrey–Clarke–Groves

PAYS

Pay the loss you cause the others

USED BY

Meta (Facebook) ads

TRUTHFUL BIDDING BEST?

Yes

GFP

Generalized First Price

PAYS

Pay your own bid

USED BY

Overture, 1997–2002

TRUTHFUL BIDDING BEST?

No

MODEL 3 · WORKED EXAMPLE

Truthful bidding is not stable under GSP

5 advertisers value a click at ₹12, 10, 8, 6, 4. 3 slots with 30%, 20%, 10% clicks.

Rule and bids	Bids (₹)	Price per click, slots 1–3 (₹)	Revenue per search	Stable?
VCG, truthful	12, 10, 8, 6, 4	8.0, 7.0, 6.0	₹4.40	Yes
GSP, truthful	**12, 10, 8, 6, 4**	**10.0, 8.0, 6.0**	**₹5.20**	**No**
GSP, equilibrium	12, 8, 7, 6, 4	8.0, 7.0, 6.0	₹4.40	Yes
GFP, truthful	12, 10, 8, 6, 4	12.0, 10.0, 8.0	₹6.40	No

The same three brands win in every case, so the value created (₹6.40 per search) never changes. Only the payments do.

BRAND A's BETTER MOVE

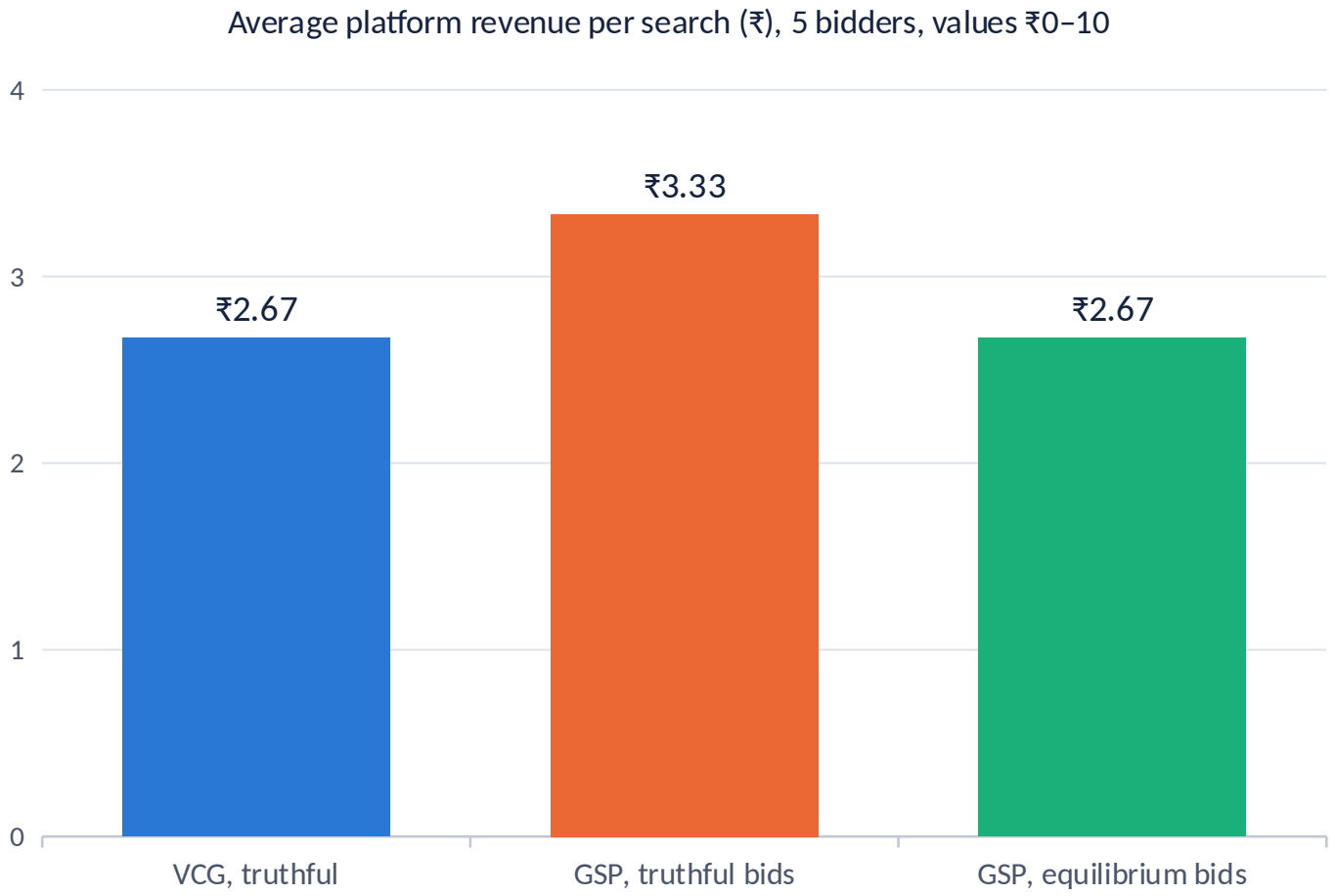
Bid ₹12 → slot 1
0.30 × (12 − 10) = ₹0.60

Bid just above ₹8 → slot 2
0.20 × (12 − 8) = ₹0.80

+33% profit by bidding lower and taking a worse slot.

MODEL 3 · 10,000 RANDOM AUCTIONS

In equilibrium, GSP earns exactly what VCG earns



Truthful GSP looks 25% richer than VCG (₹3.33 vs ₹2.67)

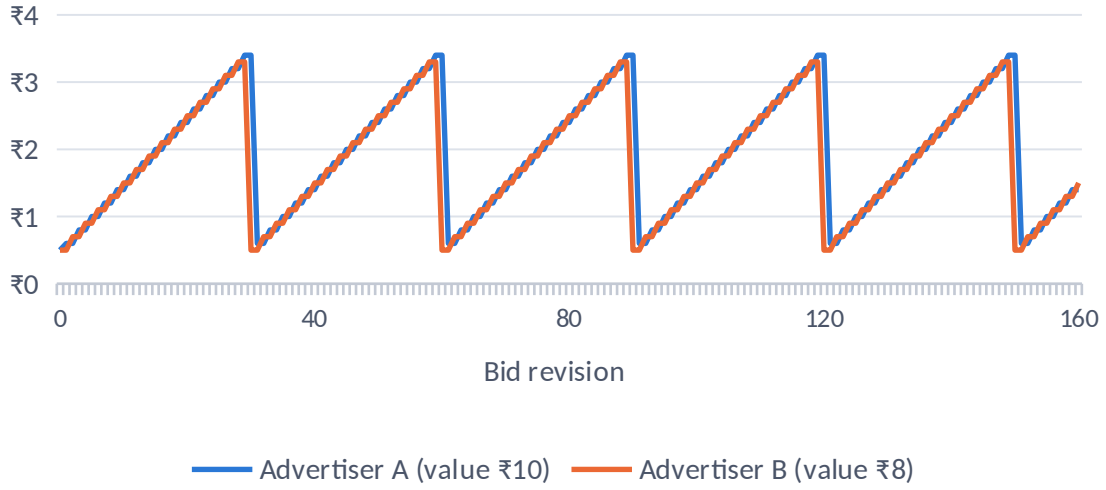
But advertisers do not bid truthfully. Once they bid strategically, GSP earns ₹2.67, the same as VCG, in every auction

Efficiency is the same: all three give ₹4.33 of value per search; only its split changes

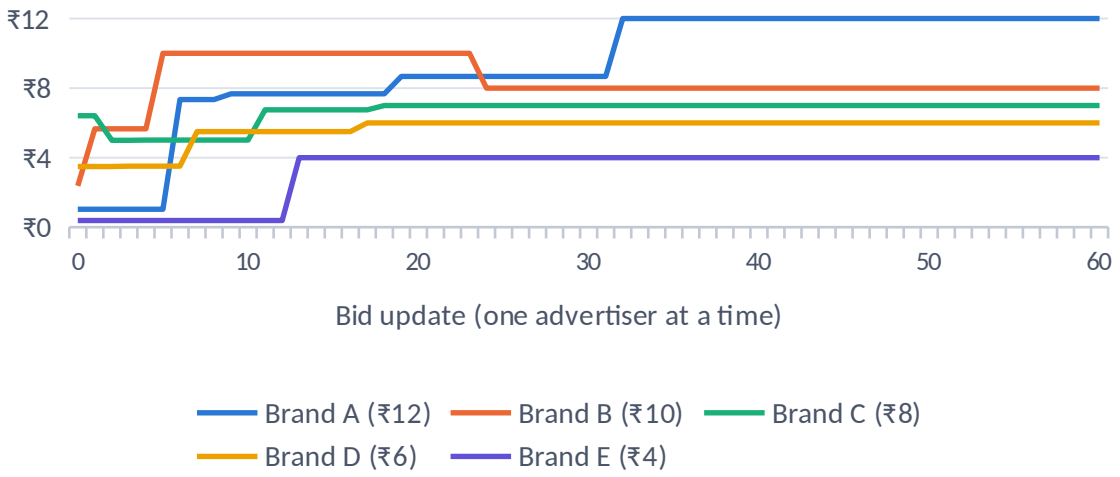
Matches theory: Edelman et al. (2007), Varian (2007)

First-price bidding never settles; GSP does

First-price (GFP): bid per click over 160 moves



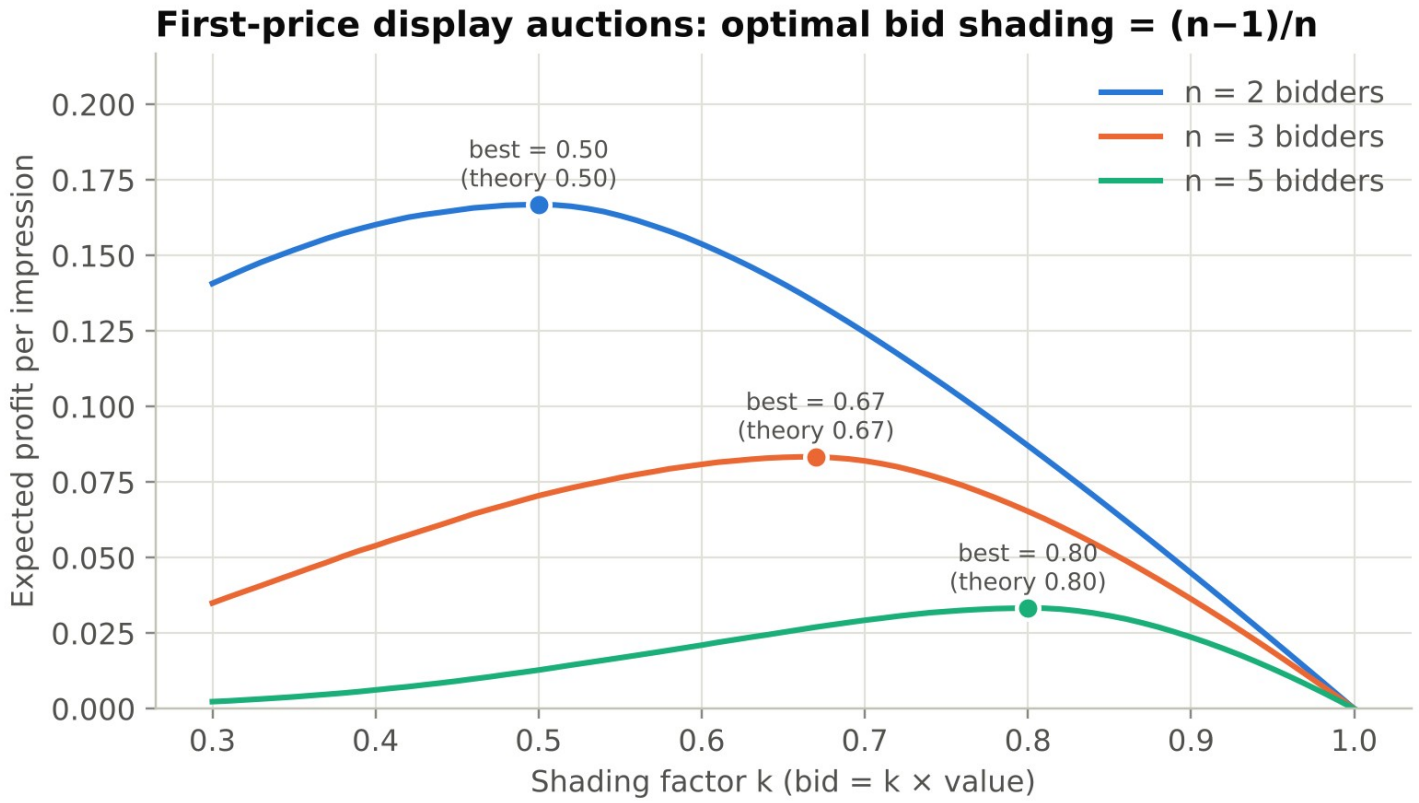
GSP balanced bidding: bids settle at 12, 8, 7, 6, 4



13 crashes in 400 bid changes. Each firm outbids the other by ₹0.10 until slot 2 at the reserve is better, then the war restarts.

20 / 20 random starts converge to the equilibrium bids in 11 to 41 updates. Revenue ends at ₹4.40, the VCG value.

Shade your bid to (n – 1) ÷ n of your value

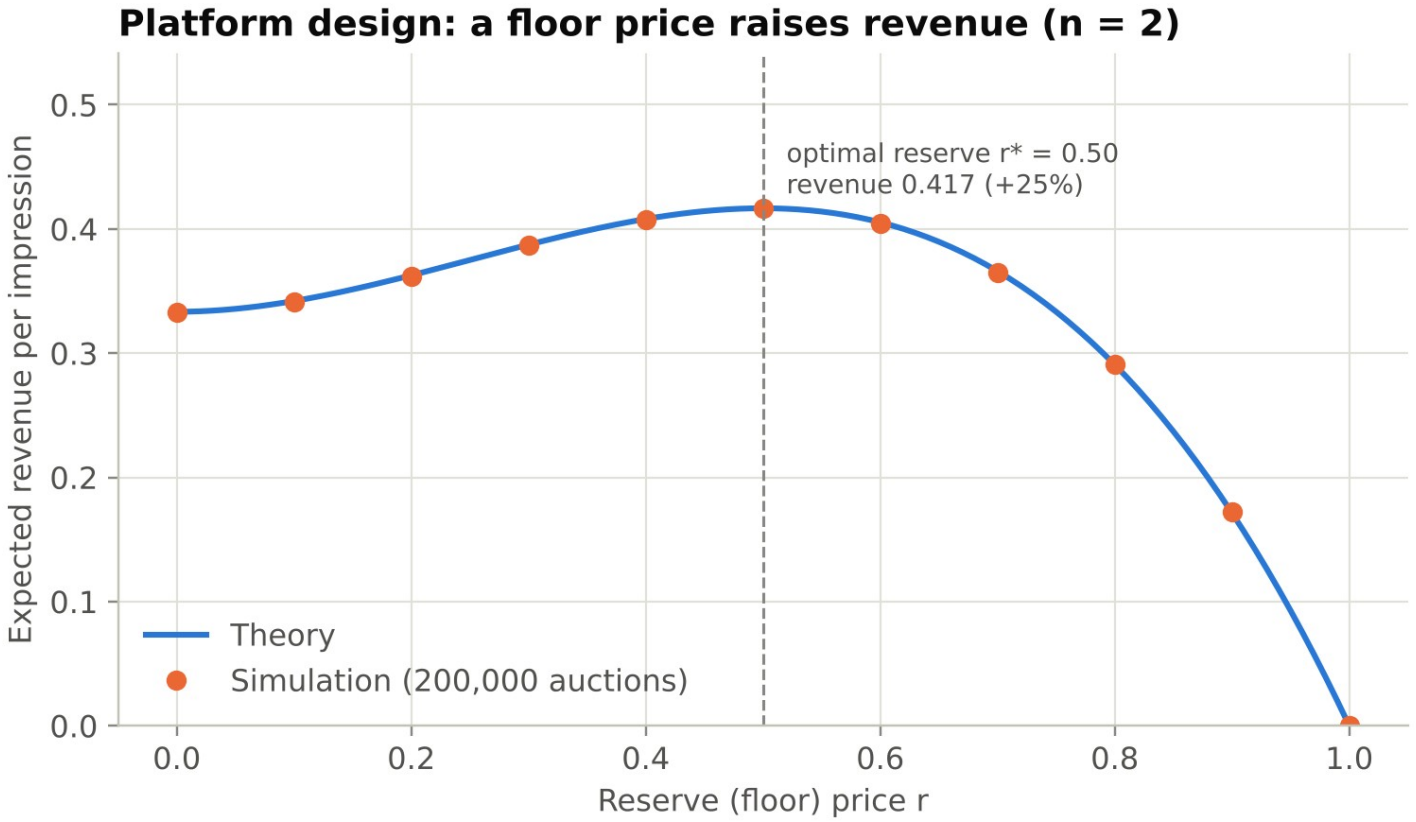


Every page view sells one ad in a first-price auction. Bidders do not know rivals' values (Bayesian game, values uniform 0–1).

Bidders	Best shading (simulated)	Theory
2	0.50	0.50
3	0.67	0.67
5	0.80	0.80

More rivals means less shading. Google Ad Manager moved to first-price auctions in 2019, so bid-shading tools became a major ad-tech product.

A reserve price of 0.5 raises revenue by 25%



+25%

revenue with the best reserve (0.333 → 0.417, two bidders)

25%

of impressions go unsold at that reserve, and it still pays

Revenue equivalence: first-price and second-price earn the same, (n − 1) ÷ (n + 1), within 0.001 for n = 2 to 10.

IMPLEMENTATION

How the software is built

MODELS

adgame/ package

- normal_form.py: NE solver
- ad_spend_game.py: Model 1
- repeated_game.py: Model 1b
- budget_contest.py: Model 2
- position_auctions.py: Model 3
- dynamics.py: Model 4
- auction_design.py: Model 5

RUNNERS

three ways to use it

- main.py: runs every experiment
- app.py: Streamlit dashboard
- tests/: 20 automated tests

OUTPUTS

what you get

- 12 result figures
- CSV tables + results.json
- Live what-if dashboard
- Test report: 20 passed

BUILT WITH

Python | NumPy | Matplotlib | Pandas | Streamlit | Pytest

All solvers written from scratch; no game-theory library computes any result.

VERIFICATION

Every result matches published theory

20 / 20

automated tests pass in under 1 second

Our mixed-equilibrium solver is also cross-checked against the nashpy library.

Model	Theory predicts	Our program finds
1 Ad war	(High, High) is the only NE	(High, High); 50% of profit lost
1b Repeated	Cooperate if $\delta \geq (T-R)/(T-P)$	$\delta^* = 0.20$; reciprocity wins
2 Contest	Total spend = $(n-1)V/n$	Exact for n = 2 to 20
3 Auctions	VCG truthful; GSP-EOS = VCG	Confirmed on 10,000 auctions
4 Dynamics	GFP cycles; GSP converges	13 crashes; 20/20 converge
5 Display	Shade $(n-1)/n$; reserve 0.5	0.50 / 0.67 / 0.80; $r^* = 0.50$

LIVE DEMO

The interactive dashboard

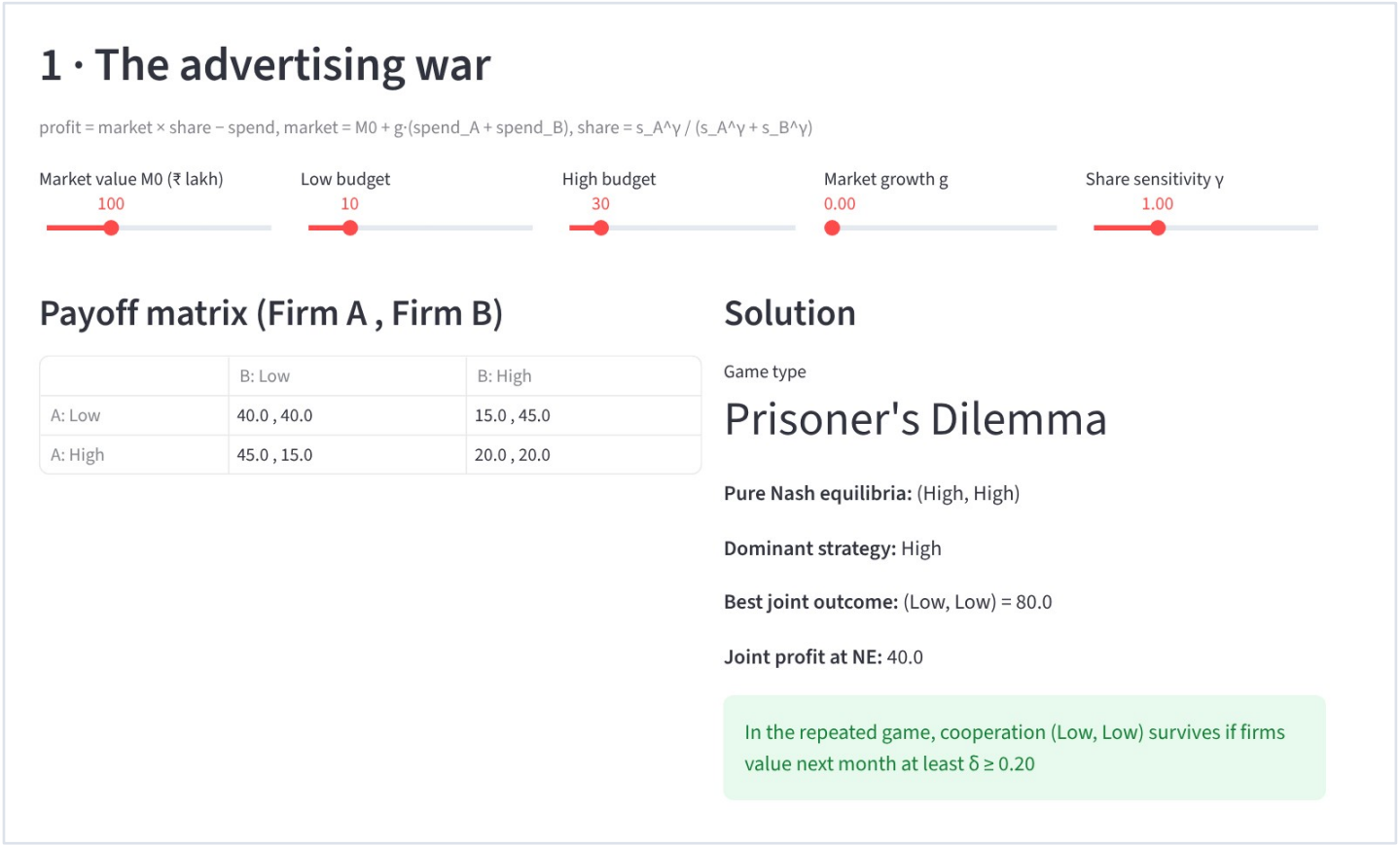
```
streamlit run app.py
```

1. **Advertising war**
   Move growth g above 1: the game changes from Prisoner's Dilemma to Deadlock

2. **Budget contest**
   Set Firm 3's cost to 3.0: its budget drops to zero

3. **Keyword auctions**
   See Brand A's profitable deviation under truthful GSP

4. **Bidding dynamics**
   Watch the first-price saw-tooth and GSP settle



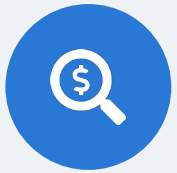
APPLICATIONS

Where these results are used



Advertisers

Spend more only when ads grow the market. Shade bids in first-price auctions. Expect crowded markets to eat your margin.



Ad platforms

Pick stable auction rules (GSP or VCG over first-price). Set reserve prices. Use quality scores in ranking.



Ad-tech firms

Build bid-shading and automated bidding tools based on best-response and balanced-bidding rules.



Regulators

Watch for tacit coordination on budgets in repeated play, and for platform power that grows with advertiser numbers.

CONCLUSION

Key takeaways

01 Budgets

Firms overspend when ads steal customers: 50% of joint profit lost with 2 firms, 90% of value to the platform with 10.

02 Repetition

Low budgets can last without any agreement if $\delta \geq 0.20$, but one mistake can break trigger strategies.

03 Auction rules

GSP is not truthful, yet in equilibrium earns exactly VCG revenue. First-price keyword bidding never settles.

04 Design

Shade bids to $(n-1)/n$; formats earn the same; a 0.5 reserve adds 25% revenue.

Limits and next steps: numbers are illustrative, not fitted to real data. Next: real auction datasets, learning bidders, and budget-limited auto-bidding agents.

THANK YOU

Questions?

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KEY REFERENCES

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- Varian (2007), Int. Journal of Industrial Organization 25(6)
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Code, report and dashboard: python main.py · python -m pytest · streamlit run app.py